

Practice Final – Solutions

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Problem 1. Consider generating a random variable X such that

$$\mathbb{P}(X = i) = \frac{e^{-\lambda} \lambda^i / i!}{\sum_{j=0}^k e^{-\lambda} \lambda^j / j!}, \quad i = 0, 1, 2, \dots, k.$$

(a) Derive the inverse transform method to generate X .

(b) Suppose you run the following algorithm to generate X : First sample from a Poisson distribution with rate λ , and if this value falls into $\{0, 1, \dots, k\}$, you output it; otherwise you return to the sampling step and repeat until you obtain an output. Is this an acceptance-rejection procedure? If it is, state precisely all the algorithmic configurations in this procedure.

Solution.

(a) We have

$$\mathbb{P}(X = i + 1) = \frac{\lambda}{i + 1} \mathbb{P}(X = i)$$

The following procedure generates X :

Step 1. Generate $U \sim \text{Unif}(0, 1)$.

Step 2. $i = 0$; $q = \frac{e^{-\lambda}}{\sum_{j=0}^k e^{-\lambda} \lambda^j / j!}$; $F = q$.

Step 3. If $U \leq F$, $X = i$ and stop.

Step 4. $q = \frac{\lambda}{i+1} \cdot q$, $F = F + q$, and $i = i + 1$.

Step 5. Go to Step 3.

(b) The algorithm suggested in the problem is given as follows;

Step 1. Generate $Y \sim \text{Pois}(\lambda)$.

Step 2. If $Y \leq k$, let $X = Y$ (accept); else go to Step 1.

Now consider the candidate function $g(x)$ as a pmf of a Poisson random variable with parameter λ , and $f(x)$ as the given pmf

$$g(x) = e^{-\lambda} \lambda^x / x!, \quad x = 0, 1, 2, \dots$$

Let $C = \left(\sum_{j=0}^k e^{-\lambda} \lambda^j / j! \right)^{-1}$, be the scaling constant. Let us rewrite the $f(x)$.

$$f(x) = C e^{-\lambda} \lambda^x / x!, \quad x = 0, 1, 2, \dots, k$$

Consider the ratio $\frac{f(x)}{g(x)}$ for any $x = 0, 1, 2, \dots, k$,

$$\frac{f(x)}{g(x)} = \frac{C e^{-\lambda} \lambda^x / x!}{e^{-\lambda} \lambda^x / x!} = C,$$

and for $x > k$, $\frac{f(x)}{g(x)} = 0$.

Therefore, $\frac{f(x)}{Cg(x)} = 1$ if $x \leq k$, and $= 0$ if $x > k$.

As you can see the ratio test against a uniform random variable, $\frac{f(x)}{Cg(x)} \geq U$, $U \sim Unif[0, 1]$, does not actually depend on the value of x or the value of C . Since it is either 1 whenever Y is less than k and we accept in this case or 0 when Y is larger than k in which case we reject. Thus, the candidate algorithm indeed applies Acceptance-Rejection method with some simplifications in the procedure.

Problem 2. Let $\alpha, \beta > 0$. Give a method for generating a random variable having distribution function

$$F(x) = 1 - \exp(-\alpha x^\beta), 0 < x < \infty.$$

- (a) Derive an inverse transform method to generate X .
(b) Suppose you insist to use an acceptance-rejection method to generate X , using a proposal that has the same distribution function as above but with a new $\tilde{\beta}$ such that $\tilde{\beta} > \beta$, and that your computer has an established routine to generate from this proposal distribution directly. Does this acceptance-rejection method work? Explain.

Solution.

- (a) $U = 1 - \exp(-\alpha x^\beta)$. Solving for X in terms of U gives;

$$1 - U = \exp(-\alpha x^\beta), \quad \log(1 - U) = -\alpha x^\beta$$

$$\frac{\log(1 - U)}{-\alpha} = x^\beta, \quad x = \left(\frac{\log(1 - U)}{-\alpha}\right)^{\frac{1}{\beta}}$$

Thus, we can generate the random variable X with the following algorithm

Step 1. Generate $U \sim Unif(0, 1)$.

Step 2. Return $X = \left(\frac{\log(1-U)}{-\alpha}\right)^{\frac{1}{\beta}}$

In fact you can also write U instead of $1 - U$ in Step 2 for a more elegant presentation.

- (b) No it does not work. Recall that to use Acceptance-Rejection method, we need to be able to find C such that

$$\frac{f(x)}{g(x)} \leq C$$

for all x . Now plug in $f(x) = F(x)x = \alpha\beta x^{\beta-1} \exp(-\alpha x^\beta)$ and $g(x) = \alpha\tilde{\beta}x^{\tilde{\beta}-1} \exp(-\alpha x^{\tilde{\beta}})$ with $\tilde{\beta} > \beta$. This pdf is defined only on non-negative half-line so we should have some C such that $\frac{f(x)}{g(x)} \leq C$ for all $x \geq 0$. Consider large x , i.e. take limit of $\frac{f(x)}{g(x)}$

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{x \rightarrow \infty} \frac{\alpha\beta x^{\beta-1} \exp(-\alpha x^\beta)}{\alpha\tilde{\beta}x^{\tilde{\beta}-1} \exp(-\alpha x^{\tilde{\beta}})} = \lim_{x \rightarrow \infty} \frac{\beta}{\tilde{\beta}} x^{\beta-\tilde{\beta}} \exp(-\alpha x^\beta + \alpha x^{\tilde{\beta}})$$

However, we see that in the term in the exponent, because $\tilde{\beta} > \beta$, this limit will blow up to infinity since the exponential term increases very fast and dominates the polynomial term. Thus, we can never find a C such that $\frac{f(x)}{g(x)} \leq C$ for all x is satisfied, because there is always a very large x that will make $\frac{f(x)}{g(x)}$ greater than C . The calculations related to the limit are as follows;

$$= \lim_{x \rightarrow \infty} \frac{\beta}{\tilde{\beta}} x^{\beta-\tilde{\beta}} \exp(\alpha x^{\tilde{\beta}}(1 - x^{\beta-\tilde{\beta}}))$$

since $x \rightarrow \infty$, for large x we have, $x^{\beta-\tilde{\beta}} \leq 1/2$ thus we can write

$$\geq \lim_{x \rightarrow \infty} \frac{\beta}{\tilde{\beta}} x^{\beta-\tilde{\beta}} \exp\left(\frac{\alpha}{2} x^{\tilde{\beta}}\right)$$

Let us also ignore constants and also divide by x^{β} which is a constant larger than 1. Then by classical calculus results, we get

$$\geq \lim_{x \rightarrow \infty} x^{-\tilde{\beta}} \exp\left(\frac{\alpha}{2} x^{\tilde{\beta}}\right) = \infty$$

Problem 3. A highway department has enough salt to handle a total of 80 inches of snowfall. Suppose the daily amount of snow has a mean of 1.5 inches and a standard deviation of 0.3 inches.

- Approximate the probability that the salt on hand will suffice for the next 50 days.
- What theorem did you use in solving part (a)? Do you think the theorem is justified? Explain briefly.
- Suppose the highway department now has the capability to adjust its amount of salt. How many inches of snowfall should it prepare its salt for so that it is enough for the next 50 days with 99.5% chance?

Solution.

- Let X_i denote the snowfall on day i , and assume $X_i \sim N(1.5, 0.3^2)$. We want

$$P\left(\sum_{i=1}^{50} X_i \leq 80\right).$$

Assuming independence of the X_i 's, by the central limit theorem

$$\frac{\sum_{i=1}^{50} (X_i - 1.5)}{\sqrt{0.3^2 \cdot 50}} \approx N(0, 1).$$

Thus

$$P\left(\sum_{i=1}^{50} X_i \leq 80\right) = P\left(\frac{\sum_{i=1}^{50} (X_i - 1.5)}{\sqrt{0.3^2 \cdot 50}} \leq \frac{80 - 75}{\sqrt{0.3^2 \cdot 50}}\right) = P(Z \leq 2.357) = \Phi(2.357) \approx 0.9908,$$

where Z is a standard normal random variable.

- The theorem used is the central limit theorem (CLT) for sums (or averages) of i.i.d. random variables.
 - We assumed that the daily amounts of snow are independent from day to day. This may not be fully justified because weather patterns tend to persist (if it snows today, it may be more likely to snow tomorrow), so strict independence can be questionable.
 - We also used a normal approximation, which is generally justified here because the sample size (50 days) is reasonably large (greater than 30).

(c) Now we want to find x such that

$$P\left(\sum_{i=1}^{50} X_i \leq x\right) = 0.995.$$

Using the same CLT approximation:

$$P\left(\sum_{i=1}^{50} X_i \leq x\right) = \Phi\left(\frac{x - 75}{\sqrt{0.3^2 \cdot 50}}\right) = 0.995.$$

From the standard normal table, $\Phi(2.576) = 0.995$, so we set

$$\frac{x - 75}{\sqrt{0.3^2 \cdot 50}} = 2.576 \quad \Rightarrow \quad x = 75 + 2.576\sqrt{0.3^2 \cdot 50} \approx 80.4645.$$

So the department should prepare for approximately 80.46 inches of snowfall.

Problem 4. Suppose that $X_1, \dots, X_n$ are normal with mean μ_1 ; $Y_1, \dots, Y_n$ are normal with mean μ_2 ; and $W_1, \dots, W_n$ are normal with mean $\mu_1 + \mu_2$. Assuming that all $3n$ random variables are independent with a common variance σ^2 , find the maximum likelihood estimators of μ_1 and μ_2 .

Solution. The joint density of the $3n$ variables is

$$f(x_1, \dots, x_n, y_1, \dots, y_n, w_1, \dots, w_n; \mu_1, \mu_2) = (2\pi\sigma^2)^{-3n/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n [(x_i - \mu_1)^2 + (y_i - \mu_2)^2 + (w_i - \mu_1 - \mu_2)^2]\right).$$

The log-likelihood is

$$\begin{aligned} \ell(\mu_1, \mu_2) &= \log f(x_1, \dots, x_n, y_1, \dots, y_n, w_1, \dots, w_n; \mu_1, \mu_2) \\ &= -\frac{3n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n [(x_i - \mu_1)^2 + (y_i - \mu_2)^2 + (w_i - \mu_1 - \mu_2)^2]. \end{aligned}$$

To maximize $\ell(\mu_1, \mu_2)$, we take partial derivatives and set them to zero:

$$\begin{aligned} \frac{\partial \ell}{\partial \mu_1} &= \frac{1}{\sigma^2} \sum_{i=1}^n [(x_i - \mu_1) + (w_i - \mu_1 - \mu_2)] = 0, \\ \frac{\partial \ell}{\partial \mu_2} &= \frac{1}{\sigma^2} \sum_{i=1}^n [(y_i - \mu_2) + (w_i - \mu_1 - \mu_2)] = 0. \end{aligned}$$

Dividing both equations by $-2n$ gives

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n x_i + \frac{1}{n} \sum_{i=1}^n w_i - 2\mu_1 - \mu_2 &= 0, \\ \frac{1}{n} \sum_{i=1}^n y_i + \frac{1}{n} \sum_{i=1}^n w_i - \mu_1 - 2\mu_2 &= 0. \end{aligned}$$

Solving this linear system yields

$$\hat{\mu}_1 = \frac{\sum_{i=1}^n w_i + 2 \sum_{i=1}^n x_i - \sum_{i=1}^n y_i}{3n}, \quad \hat{\mu}_2 = \frac{\sum_{i=1}^n w_i + 2 \sum_{i=1}^n y_i - \sum_{i=1}^n x_i}{3n}.$$

Problem 5. A sample of 20 cigarettes is tested to determine nicotine content and the average value observed was 1.2 mg. You may assume that the nicotine content of each cigarette is normally distributed, and that the standard deviation of a cigarette's nicotine content is $\sigma = 0.2$ mg.

- (a) Compute a 95% confidence interval for the mean nicotine content of a cigarette.
- (b) Find the number of observations needed, in addition to the 20 already obtained, to ensure the point estimate is within ± 0.01 mg from the ground-truth mean nicotine content with 95% confidence.
- (c) Are your answers in parts (a) and (b) well justified by the assumptions? Explain briefly.

Solution.

- (a) From the standard normal table, $z_{0.975} = 1.96$. A 95% confidence interval for the mean is

$$\left(1.2 - 1.96 \cdot \frac{0.2}{\sqrt{20}}, 1.2 + 1.96 \cdot \frac{0.2}{\sqrt{20}} \right) = (1.1123, 1.2877).$$

- (b) The total required sample size is

$$n \geq \left(z_{0.975} \cdot \frac{\sigma}{\epsilon} \right)^2 = \left(1.96 \cdot \frac{0.2}{0.01} \right)^2 = 1536.64$$

so the total required sample size is 1537. The additional sample size is $1537 - 20 = 1517$.

- (c) They are well-justified. Since the population is normal, the confidence interval and the required sample size in parts (a) and (b) are always valid.